ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech. (CSE-CC)

Semester: IV

SUBJECT CODE: BCSICT401

INFORMATION SECURITY

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. **(10 Marks)**

Q. 1. Define the following terms: **(1 × 5 = 5 Marks)**

(i) What is meant by information security?
(ii) Define risk in the context of information security.
(iii) What is declassification?
(iv) What is traffic confidentiality?
(v) What is the purpose of RSA algorithm?

Q. 2. Answer the following questions: **(1 × 5 = 5 Marks)**

(i) What is meant by information security?
(ii) Define risk in the context of information security.
(iii) What is a security attack?
(iv) What do you understand by intruders?
(v) Define hackers.

SECTION – B

NOTE:Attempt all questions. **(5×2×2 = 20 Marks)**

Q.3.(i)(a) Explain the different types of security policies at Tier 1, Tier 2, and Tier 3 levels.

OR

(b) Describe the steps involved in developing a risk management program.

(ii)(a) Describe the key components of user identity and access management.

OR

(b) What is the function of intrusion detection systems and event logging in network security?

Q.4.(i)(a) Describe the structure and functioning of IP Security (IPsec) in securing data at the network layer.

OR

(b) What are the key steps in key management for secure communication?

(ii)(a) Describe the architecture and complete working of DES algorithm.

OR

(b) Explain in detail the functioning of Pretty Good Privacy.

SECTION – C

NOTE: Attempt all questions. **(10 × 2 = 20 Marks)**

Q.5.(a) Describe the Data Encryption Standard (DES) in detail, including its structure, key generation, and encryption process. Compare its security with AES.

OR

(b) Describe the components and working of physical and logical access control systems. Include perimeter security, intrusion detection, and account privilege management in your explanation.

Q.6.(a) Explain RSA algorithm with an example. Discuss key generation, encryption, and decryption processes. How does RSA ensure security in public key cryptography?

OR

(b) Elaborate on the different stages of key lifecycle management in a secure system. Explain how key generation, usage, update, and destruction are handled to maintain confidentiality.